How a Plane Gets Into the Sky and Stays There

Have you ever watched a plane disappear into the clouds and wondered how something so heavy can fly? A jumbo jet can weigh over 400 tonnes, yet it climbs into the sky with ease. Planes are extraordinary machines, and the science behind them is fascinating. Read on to discover how three key parts work together to lift a plane off the ground and keep it soaring.

Wings

A plane's wings are its most important feature. As the engines push the plane forward, air flows over and under each wing. Due to the special curved shape of the wing, air moves faster over the top than underneath. As a result, this difference in air pressure creates a powerful upward force called lift, which pushes the plane into the sky.

Engines

Without thrust, a plane cannot move fast enough to take off. Jet engines work by sucking in air at the front, mixing it with fuel and burning it. By doing this, the engine produces a powerful blast of hot gas that shoots backwards. As a result, the plane is pushed forwards at tremendous speed, which gives the wings enough airflow to generate lift.

The Tail

Once a plane is in the air, the pilot needs to control its direction. The tail contains moveable surfaces called rudders and elevators. By adjusting these surfaces, the pilot can steer left or right and climb or descend. Therefore, the tail is what allows the pilot to navigate safely through the sky and bring the plane in to land.

Planes are remarkable machines. However, by working together, the wings, engines and tail make flight possible. Whether you are watching from the ground or sitting in a window seat, these three parts are working constantly to keep the plane safe in the sky.

Annotation key — use four colours

Colour 1: Rhetorical question
question that draws the reader in without needing an answer

Colour 2: Subheadings
one-word label that organises each section

Colour 3: Causal connectives
therefore, as a result, due to, since, which means

Colour 4: Fronted adverbials
By [doing this]… / As a result,… / When this happens…

Annotation notes